

20250721EB_MN (ATP, NCT, TDP43, IF)

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Status: In progress
Entry Created On: 06 Aug 2025 18:40:10 UTC
Entry Last Modified: 15 Sep 2025 19:34:34 UTC
Export Generated On: 01 Sep 2026 18:27:18 UTC

- 2025年7月21日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年7月23日 星期三
- T2 -> T3
- 2025年7月25日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年7月28日 星期一
- T4 -> T5
- 2025年7月30日 星期三
- T5 -> T6
- 2025年8月1日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年8月4日 星期一
- T6 -> T6
- 2025年8月5日 星期二
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs

2. Remove the PLO and wash the plate with PBS once

3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add 2mL accutase in 15mL falcon

6. Put the tube into 37C water bath. Flick the cells every 5 min until the accutase went cloudy (~45min this time). (There are many cells attached on the well of 15mL falcon. Will use 50mL next time)

7. pipette the accutase gently and slowly around 20-30 times to break the EBs by checking if there are still visible EBs. (I can still see a lot of undissociated EBs this time, so I take the accutase without visible EBs for the further process.
8. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.
9. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
10. Filter the cell with 40um filter into a new 50mL tube
11. Spin the cell down at 100rcf, 3.5min. Remove the supernatent.
12. Add 1ml of T7 to resuspend the cells thouroughly to avoid clusters.
13. Add more T7 (~5mL) to resuspend the cells and count it.
14. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	P2	6

Table1 ^

	A	B
1	run1	volume (mL)
2	accutase	2
3	HI-FBS/CTWM	4

Table3 ^

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	3.12	92	5	15.6	A	24	225	24	1731

Well1 ^

	1	2	3	4	5	6
A						
B						
C						
D	92 5uL	110 15uL	111 24uL			

- 2025年8月6日 星期三

T7-> T7
- 2025年8月8日 星期五

T7 -> T8
- 2025年8月10日 星期日

1/2 T8 -> 1/2 T8
- 2025年8月12日 星期二

1/2 T8 -> 1/2 T8
- 2025年8月13日 星期三

Image NCT, ATP

Signal of 111 improved, but the intensity still very low

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

^

ATP O.C. , 60X, 600x900, 16-bit				
	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	20	200
3	Dapi-ex-Green-em	16	20	300

^

Treatment time (8/13/2025)								
	A	B	C	D	E	F	G	H
1			110		111		92	
2	#		treat time	image time	treat time	image time	treat time	image time
3	1	2.5mM glu	12:30	13:30	10:17	11:17	15:10	16:10
4	2	0.15mM glu	13:10	14:09	10:46	11:49	15:25	16:25
5	3	0mM glucose 1h	13:50	14:50	11:15	12:15	15:49	16:49
6	4	0mM glucose +FCCP+2DG 1h	14:28	15:29	11:45	12:45	16:00	17:05

Treat SPY595 (1:3000) for 4AB and 5AB
take 100ul media BP and add 0.33 uL of SPY595. Treat each well with 50uL

2025年8月14日 星期四

Imaging TDP43
Tunicamycin 5mg/mL = 6.12 mM, 100 µg/mL = 122 uM
or 0.1uM
dilute 1mM JFX650
--> 1:400 --> 2.5uM
--> 1:2500 --> 1nM (working concentration)

Table4							
	A	B	C	D	E	F	G
1		stock (mM)	working (uM)	dilution	1st dilution	2nd dilution	800uL in total
2	MG132	10.5	2	5250	52.5	100	8.0
3	Tunicamycin	6.12	1	6120	61.2	100	8.0
4	Thapsigargin	1	20	50	1	50	16.0
5	arsenite	500	500	1000	1	1000	0.8

FOR IF

Well2 ^

	1	2	3	4	5	6
A	1	1	1	5	6	1
B	2	2	2	7	8	2
C	3	3	3	1	1	1
D	4	4	4	2	2	2

O.C. , 60X, 600x900, 16-bit1 ^

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	20	300ms	
3	640	16	10	300ms	

Table2 ^

	A	B
1	condition#	condition
2	1	2.5mM glucose 1h
3	2	0.15mM glucose 1h
4	3	0mM glucose 1h
5	4	0mM glucose +FCCP+2DG 1h
6	5	2.5mM glucose + DMSO
7	6	2.5mM glucose +2uM MG132
8	7	2.5mM glucose + 1uM tunicamycin
9	8	2.5mM glucose +500uM arsenite

Treatment time (all)1										
	A	B	C	D	E	F	G	H	I	J
1			TDP43							
2	#		treat JFX	image time 1h	image time 2h	image time 3h	image time 4h	image time 5h	image time 6h	image time 7h
3	5	2.5mM glucose + DMSO								
4	6	2.5mM glucose +2uM MG132								
5	7	2.5mM glucose + 1uM tunicamycin								
6	8	2.5mM glucose +500uM arsenite								
			11:20-11:25	12:15-25	13:20-13:30	14:20-14:40	15:20	16;20	17:20	18:20

2 hours after arsenite, the nuclear shrink and cell became round.
imaging order: 5 -> 6 -> 8 -> 7

treat the rest of cells at 17:50
Wash cells with cold PBS and fix cells (columns 4-6) with 4%PFA at 19:50
wash with PBS
0.2% triton for 15min
PBS wash
10% FBS --> cold room over weekend

2025年8月18日 星期一

Treat cells with primary antibody

IF antibody						
	1	2	3	4	5	6
A				A	A	B
B				A	A	B
C				C	D	E
D				C	D	E

Table7

	A	B	C	D	E	F	G
1	condition	Ab#	antibody	species	dilution	total volume	
2	A	33	TDP-43	Rb	500	800	1.6
3	B	33	TDP-43	Rb	500	400	0.8
4	B	C.2	G3BP1	m	500	400	0.8
5	C	109	phospho TDP-43 (S409/410)	Rb	500	400	0.8
6	C	C-34	Orf1p	m	1000	400	0.4
7	D	66	Lamin B1 Nuclear envelope marker	Rb	1000	400	0.4
8	D	C-34	Orf1p	m	1000	400	0.4
9	E	C-31	Rangap1	Rb	500	400	0.8
10	E	C-34	Orf1p	m	1000	400	0.4

2025年8月19日 星期二

2nd antibody & DAPI staining

2025年8月20日 星期三

Imaging IF

A4,5 & B4,5 OC 60x,600x900, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	640	30	300	
3	561	30	100	very dim
4	488	20	100	
5	z-stack	1um/step	17 steps in total	

The rest of wells

OC 60x,600x900, 16bit					^
	A	B	C	D	
1		laser power (%)	exposure time (ms)		
2	405	10	100		
3	488	20	100	very dim	
4	561	30	300		
5	z-stack	1um/step	17 steps in total		